

COMPUTER SCIENCE TRIPOS Part IB 75%, Part II 50% – 2020 – Paper 7

2 Economics, Law and Ethics (ah793)

You have an idea for an innovative new type of application for self-directed teaching and learning. The *eTeacher* uses state-of-the-art natural language processing approaches to allow for text and speech interaction. Over time, it builds a profile of the user's most effective means of learning, and tailors the teaching style accordingly. The intended users are secondary school students and adults who wish to refresh their understanding of Mathematics, English, Geography, or History. You plan to monetise the application by way of advertisements and in-app purchases.

Market failure,
Principles of law

- (a) You are seeking investors. Discuss the economic and legal considerations that might inform your business plan and persuade investors to fund your startup. For a convincing pitch, you should define the necessary terms and explain your reasoning. [7 marks]

Answer: From a legal perspective, they will want to know that there is some intellectual property ownership—that you own the copyright for the code. They will need to be convinced that this has some defensible market value (2 marks).

From an economic perspective, there is a race for market share when there are new products and services. Those that have high fixed costs and low marginal costs, significant switching costs due to technical lock-in, and network effects are likely to become dominant. Once established, it can cost a lot for someone else to compete for market share (1 marks).

This app has high fixed cost (the cost to develop the technology), but low marginal costs (the cost per extra user is very low). The app can go from hundreds to thousands of users with very little extra effort (1 mark).

As the application adapts to the individual user, the cost to switch to a competitor is higher. The time investment spent with the application would be lost (1 mark).

Network effects - networks become more valuable the more people use them. When you have x people on a network, the value of the network is x squared. Additional benefit for the application is more training data to improve the accuracy of the NLP and effective learning styles. Potential ways to create networks, such as having friends and classmates take courses together (2 marks).

Market failure

- (b) Explain how things might change if the economy goes into recession. [3 marks]

Answer: In times of recession liquidity preferences rise. People cut back on spending and start saving, so may be less likely to click on advertisements for products. Banks require more in reserve, so call in loans and limit borrowing. This may make it more difficult to attract investors (1 mark). Fewer jobs may be available, therefore people may be unwilling to invest in furthering their education or using your app (although on the other hand, they may increase their investment in an effort to retain/gain employment) (1 mark). Investors may be unwilling to fund the technology industry due to concerns about the bubble bursting (recessions are often linked to technological change) (1 mark).

Law and the
Internet

- (c) Outline the principles of the General Data Protection Regulation (GDPR). How might you comply with the GDPR and allow individuals to exercise their rights? [10 marks]

Answer: The application allows users to interact by speech. Voices are considered personal data under the GDPR. Also likely to collect other data, such as when users register. Users may also disclose personal data during their interaction with the application (1 mark).

There are six principles to be complied with, namely that data are:

- (i) fairly and lawfully processed
- (ii) processed for limited purposes
- (iii) adequate, relevant, and not excessive
- (iv) accurate and up to date
- (v) not kept in a form that identifies people for longer than necessary
- (vi) processed securely and protected against loss or damage (1 mark)

You should implement appropriate technical and organisational measures to demonstrate you comply with the GDPR. These may include internal data protection policies such as staff training, internal audits of processing activities, and reviews of internal HR policies (1 mark).

You should maintain relevant documentation on processing activities. Measures to meet the principles of data protection by design and default include data minimisation, pseudonymisation, transparency, allowing the user to monitor processing, and creating and improving security features on an ongoing basis. You should design data protection into new systems and do an impact assessment. As you will be processing data at scale you should appoint a Data Protection Officer. Data breaches must be reported to the regulator within 72 hours (2 marks).

You should identify why processing is allowed. One way is by seeking consent to process the data. Consent must be freely given, specific, informed and unambiguous. You cannot use pre-ticked boxes or infer it. As this application will be used by schoolchildren, you will need to consider the age of consent. For those in the UK, you must get permission from parents for children under 13, however elsewhere in Europe the age limit is 16. May also be able to claim legitimate interest. (2 marks).

To comply with the rights for individuals, you should:

- The right to be informed: provide a notice that explains your processing
 - The right of access: design your system so this is possible
 - The right to rectification: correct errors
 - The right to erasure: allow for data to be erased if there is no compelling reason for retention
 - The right to restrict processing: allow for the data to be kept but not otherwise processed
 - The right to data portability: allow for data to be ported on request
 - Rights in relation to automated decision making and profiling: explain that you are automating decisions about learning styles, how these decisions are made, and their significance and consequences (3 marks)
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